

TECHNIQUE MASTERING EXERCISES

M DIFFERENTIAL EQUATIONS

Find the solution of each of the following differential equations with initial values.

1. $\frac{dy}{dx} = \frac{x}{2}, \quad y\left(\frac{1}{2}\right) = -1 :$

$$y = \int \frac{x}{2} dx = \frac{1}{2} \frac{x^2}{2} + c$$

$$y\left(\frac{1}{2}\right) = -1 \Rightarrow \frac{1}{4} \left(\frac{1}{2}\right)^2 + c = -1 \Rightarrow y = \frac{x^2}{4} - \frac{17}{16}$$

2. $\frac{dy}{dt} = -\frac{3t^2}{2}, \quad y(-1) = -\frac{1}{2} :$

$$y = \frac{-3}{2} \int t^2 dt = \frac{-3}{2} \frac{t^3}{3} + c = -\frac{1}{2} t^3 + c$$

$$y(-1) = -\frac{1}{2} \Rightarrow -\frac{1}{2} (-1)^3 + c = -\frac{1}{2} \Rightarrow y = -\frac{1}{2} t^3 - 1$$

3. $\frac{dy}{dx} = \sin x, \quad y\left(\frac{\pi}{2}\right) = 3 :$

$$y = \int \sin x dx = -\cos x + c$$

$$y\left(\frac{\pi}{2}\right) = 3 \Rightarrow -\cos\left(\frac{\pi}{2}\right) + c = 3 \Rightarrow y = -\cos x + 3$$

4. $\frac{dy}{dx} = e^x, \quad y(0) = 4 :$

$$y = \int e^x dx = e^x + c$$

$$y(0) = 4 \Rightarrow e^0 + c = 4 \Rightarrow y = e^x + 3$$

5. $\frac{dy}{dx} = \frac{1}{x}, \quad y(1) = 2 :$

$$y = \int \frac{1}{x} dx = \ln|x| + c$$

$$y(1) = 2 \Rightarrow \ln|1| + c = 2 \Rightarrow y = \ln|x| + 2$$

6. $\frac{d^2y}{dx^2} = 0, \quad \frac{dy(2)}{dx} = 1, \quad y(2) = 2 :$

$$\frac{dy}{dx} = \int 0 dx = c$$

$$\frac{dy(2)}{dx} = 1 \Rightarrow c = 1 \Rightarrow \frac{dy}{dx} = 1$$

$$y = \int 1 dx = x + c$$

$$y(2) = 2 \Rightarrow 2 + c = 2 \Rightarrow y = x$$

7. $\frac{d^2y}{dx^2} = \cos x$, $\frac{dy(0)}{dx} = 1$, $y(0) = 2$:

$$\frac{dy}{dx} = \int \cos x dx = \sin x + c$$

$$\frac{dy(0)}{dx} = 1 \Rightarrow \sin 0 + c = 1 \Rightarrow \frac{dy}{dx} = \sin x + 1$$

$$y = \int (\sin x + 1) dx = -\cos x + c$$

$$y(0) = 2 \Rightarrow -\cos 0 + c = 2 \Rightarrow y = -\cos x + 3$$

8. $\frac{d^2y}{dt^2} = e^t$, $\frac{dy(1)}{dt} = e$, $y(1) = -4e$:

$$\frac{dy}{dt} = \int e^t dt = e^t + c$$

$$\frac{dy(1)}{dt} = e \Rightarrow e^1 + c = e \Rightarrow \frac{dy}{dt} = e^t$$

$$y = \int e^t dt = e^t + c$$

$$y(1) = -4e \Rightarrow e^1 + c = -4e \Rightarrow y = e^t - 5e$$